

TEAM H13A-DATE · COMP3900

Human Controller Installation Manual

Webcam-based motion mapping for accessible rehabilitation gaming.

Native Desktop App | Windows & macOS | Client Handover

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01 Overview

Human Controller is built using **Tauri**, a framework for native desktop applications that combines a Rust backend with a React and TypeScript frontend. It requires hardware access (a webcam) and operating-system-level inputs, so a detailed manual installation is provided below.

Human Controller has been tested on and works for **Windows** and **macOS**. No API keys, environment variables or secrets are used, since Human Controller's database is embedded entirely in SQLite.

Installing the Application

TWO OPTIONS

RECOMMENDED FOR MOST USERS

Option A · Install the ready-built application

1. Download the installer for your system: **Human-Controller.exe** for Windows, or **Human-Controller.dmg** for macOS.
2. Open the downloaded file and follow the on-screen prompts to install.
3. Launch Human Controller like any other application, from the Start Menu on Windows or the Applications folder on macOS.
4. The first time it runs, your computer will ask for permission to use the camera. Select **Allow**.

Option B · Run from source

This option is for running Human Controller directly from its source code.

Prerequisites

Install the following before setup:

Node.js 24 (LTS)	The JavaScript runtime. Install using the official Node.js installer.
Rust (stable)	Install using the rustup installer from rustup.rs.

Windows only

- Visual Studio Build Tools, with the "Desktop development with C++" workload selected. This is required for the Rust and Tauri build toolchain.

macOS only

- Xcode Command Line Tools. Run this command:

```
xcode-select --install
```

Option B · Setup & Running

FROM SOURCE

Installation Steps

1. Clone the repository:

```
git clone https://github.com/unsw-cse-comp99-3900/capstone-project-26t2-3900-h13a-date.git
```

```
cd capstone-project-26t2-3900-h13a-date
```

2. Install the dependencies. Please use **npm ci**, not npm install, so that the exact locked versions are installed:

```
npm ci
```

Running the Application

To run the application:

```
npm run tauri dev
```

The first launch will take a few minutes while the backend compiles. Later launches are much faster. A desktop window will open automatically.

Producing a Distributable Build

To produce a distributable or installable build of the application:

```
npm run tauri build
```

Hardware and Permissions

- A **webcam** is required, since Human Controller tracks the user's body movements to control gameplay.
- The first time you run the app, your computer may ask for permission to access the camera. Please select **Allow**.